

Exploring School Engagement and Academic Expectations Using LLM-Assisted Wrangling

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Abstract—This paper presents a project proposal for incorporating Large Language Models for into data wrangling tasks using a 2012 NYC public school survey dataset focused on engagement and academic expectations.

I. INTRODUCTION

Public education in New York City is very different compared to other regions across the United States. With its competitive academic programs, diverse student population, and prominent institutions, NYC schools operate under high expectations from students, staff, and families alike. Understanding how these benchmarks are being met is essential to supporting equitable educational outcomes in NYC. By analyzing data collected through a education survey, this project aims to evaluate how well public schools are aligning with the academic and social expectations of their communities, and what factors influence students achievements.

II. RESEARCH TOPIC AND MOTIVATION

This project is motivated by the need to explore how the involvement of student’s community such as parents, teachers, and students, support achieving academic expectations.

Academic expectations refer to the benchmarks that schools believe students can and should achieve. They are usually set by the school district or state and they play a critical role in shaping student outcomes as well as financing and even accreditation.

The findings may help schools and policymakers better understand the conditions that promote and show areas of improvement.

Understanding how aspects of school climate and community engagement relate to these benchmarks is essential. This project analyzes survey data to examine these benchmarks and how to get NYC Schools to meet them. The goal is not only to measure performance but to better understand the environment that supports it.

III. RELATED WORK

There are a few related works when it comes to the topic of utilizing Large language models for data wrangling tasks.

Narayan et al. [1] pioneers using LLM’s for structured data and its accompanying data wrangling tasks. Specifically the authors highlight the use of foundation models for data

preprocessing and wrangling tasks such as data imputation and entity matching. They also look into ablation and prompt effectiveness to prove their theory that foundation models can meet SoTA standards for structured data projects.

Jaimovitch-López et al. [2] focuses on how to incorporate large language models into the preexisting data processing pipelines. An interesting advantage they show is that the need of user involvement exponentially decreases when using LLMs. While they highlight the benefits the authors also warn about the reliability of these models have to be closely watched, as the chances of errors will increase when adopting this model.

Liu et al. [3] presents a new system that is actually designed for the purpose of automating data wrangling processes. This new system is called AutoDW. What sets the system apart is its sophisticated prediction engineering and Feature Type Inference (FTI), both of which streamline the data preparation process by reducing reliance on human input.

IV. DATASET DESCRIPTION AND SOURCES

The dataset that will be used for this project is titled “2012 NYC General Education School Survey”. The data was found from the resource Data.gov, The Home of the U.S. Government’s Open Data. Upon further inspection of the dataset, it has 10 features and 1,669 observations. The data set gives a few different components that can be used to tell a really interesting story. It gives the name of the school and the type of school it is classified as such as High School or Middle School. The next part of the story is the survey response rates of different people engaged in the schools culture such as students, parents, and faculty. The data also gives the surveys scores in different aspects of schools benchmarks such as safety, communication, and academic expectations. This data can be used to find helpful conclusions in order to make recommendations for these schools.

V. PLANNED LLM IMPLEMENTATION APPROACH

To align with the projects requirement of incorporating LLM-assisted data wrangling, this project will utilize a Large Language Model (LLM) such as GPT-4o to streamline and enhance key preprocessing tasks. While the analytical techniques will be performed using traditional tools (e.g., R or

Python), the LLM will support human decision-making during data preparation in the following ways.

A. Data Cleaning and Standardization

The first way the LLM will assist is in the preprocessing tasks. The LLM will play a key role in the preprocessing stage of the project, particularly in data cleaning and standardization. It will assist in identifying and handling missing values, ensuring they are addressed consistently to improve the accuracy and reliability of the analysis. Additionally, the LLM will support the creation of a comprehensive data dictionary by interpreting and explaining each feature in the dataset. This will provide a clearer understanding of the data and make it easier for others to interpret and use the dataset effectively.

B. Data Normalization

Several fields in the dataset, such as response rates, are currently represented as percentage strings. The LLM will be used to guide the process of converting these values into appropriate numerical formats, making them more suitable for analysis. This step will involve identifying all relevant columns, recommending the most accurate normalization approach, and verifying the results for consistency.

C. Feature Engineering Guidance

The LLM will provide input on possible feature engineering strategies. This includes recommending the creation of variables and suggesting transformations that could enhance the interpretability and performance of the model. The LLM may also help identify redundant or low-variance features that could be excluded to streamline the analysis.

In addition, the LLM will help to standardize the categories that each school falls into, like whether the school is a high school or middle school. This will help to further understand engagement levels in NYC schools.

VI. PRELIMINARY DATA QUALITY ASSESSMENT

A preliminary assessment of the dataset was conducted to evaluate its overall structure, completeness, and suitability for analysis. The dataset contains information from the 2012 NYC General Education School Survey and includes 1,668 records across 10 variables. Each record represents a public school in New York City, and the variables capture response rates from parents, teachers, and students, as well as perception scores related to safety, communication, engagement, and academic expectations.

The findings of the data show a few things that the LLM can help to mitigate for the project. These are structural issues, missing values, formatting, duplicates, anomalies or outliers, and mismatching data types. These are all items that can be improved upon by the help of the LLM.

VII. SUMMARY OF PLANNED WORK

In summary, this project will provide valuable insights into public school benchmarks and engagements, allowing for better understanding of the data and potential paths forward. The current plan is to utilize the LLM in multiple ways

during the project, such as Data Cleaning and Standardization, Data Normalization, and Feature Engineering Guidance. As the project progresses further developments may be made on how to utilize the LLM and explore new ways to incorporate it.

In addition, I would like to utilize the data by conducting correlation and comparative analyses to uncover meaningful relationships within the data. Specifically, I plan to investigate disparities between different school districts and school types to identify which groups of schools may be falling short of key academic benchmarks. This will help highlight patterns and areas in need of targeted improvement. I will also view how student, staff, and parent participation affects students hitting their academic benchmarks.

Some of the questions that are proposed are: Are there significant discrepancies in the rates at which elementary, middle, and high schools meet key academic benchmarks? Is community involvement a key factor in answering the previous question? Which benchmark is the most challenging to hit? Is there correlations between school who are missing or hitting certain benchmarks? Does parent or staff involvement affect how students hit benchmarks?

To address these questions, I will use a combination of LLM-powered tools and statistical techniques within R Studio to perform both correlation and comparative analyses. The LLM will assist in shaping analytical strategies, identifying relevant features, interpreting patterns, and offering contextual insights that might otherwise be overlooked. R Studio, on the other hand, will serve as the primary environment for conducting data-driven analysis—leveraging its powerful libraries for statistical modeling, data visualization, and hypothesis testing.

By integrating these tools, I aim to produce a comprehensive analysis that not only identifies whether certain types of schools are consistently falling behind, but also uncovers the underlying factors contributing to these disparities. This includes evaluating how community engagement—measured through student, staff, and parent participation—correlates with academic outcomes. Through this approach, I hope to surface actionable insights that could support targeted interventions, resource allocation, and informed policy decisions aimed at improving educational equity and performance across different school systems.

REFERENCES

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